ANALYSIS PDCA

REPRODUCE THE DEFECT

WHAT WAS DONE TO REPRODUCE THE CONCERN? [1]	1.更新打包工序桌面5S标准、更换破损的打包桌面改为玻璃台面，更新包材的清理及检查 Update the 5S standards for the packaging workbench, replace the damaged packaging workbench with a glass surface, and update the cleaning and inspection procedures for packaging materials. 2.更新手工锡焊工序作业指导书，一次只能拿取一个产品作业，在该区域增加遮蔽防护罩，待焊接的产品放置与防护罩外面。 Update the operation instruction manual for manual soldering process. Only one product can be handled at a time. Add a shielding cover in this area. Place the product to be soldered outside the shielding cover. 3.锡焊作业指导书中增加手套跟换频次，Increase the frequency of glove changing in the soldering operation manual
HOW WAS IT DONE? [2]	模拟手工锡焊作业，待焊接产品放置与工装附近，检查待焊接产品是否有飞溅到松香异物 Simulate manual soldering operation, place the product to be welded near the fixture, and check if there are any splashes of rosin foreign objects on the product to be welded
WHEN WAS IT DONE? [3]	2025、11、7
WHAT WAS THE RESULT? [4]	刷架上有松香飞溅，状态与NG件类似 There are splashes of rosin on the brush holder, similar to the condition of NG parts

ACTION PLAN FOR PERMANENT COUNTERMEASURES - OCCURRENCE

ROOT-CAUSE [5] select the Root Cause among the list proposed	TYPE OF COUNTERMEASURE [6]	COUNTERMEASURE [7]	STANDARD NUMBER REVISION LEVEL [8]	WHO [9]	Due date [10]	Done date [11]	CHECK: Are ALL users including their Direct Manager trained to use this STANDARD? [12]	CHECK: Is there an ENFORCEMENT plan for the Direct manager to ensure the respect of this STANDARD? [13]	CHECK: Genba Check Done and OK (Date) [14]	ACTI: Document updated? Design FMEA [15]	ACTI: Document updated? Process FMEA [16]	ACTI: Document updated? Control Plan [17]	ACTI: Date when all relevant FMEAs and Control Plans were updated [18]	ACTI: Generalization to other lines and/or products within the site [19]	ACTI: LLC Done? YES /NO [20]
桌面破损未进行维护更换 The desktop was damaged and not maintained or replaced	Update an existing standard	更新各生产工序5S标准书，增加开班前对桌面灰尘、异物清洁作业 Update the 5S standard books for each production process and add the cleaning of dust and foreign objects on the desks before the start of each shift		Kerwin	Nov 2, 2025	Nov 2, 2025	YES	YES		N/A	N/A	N/A			
5S拟定不合理，未识别潜在异物混入风险 The 5S formulation was unreasonable and failed to identify the potential risk of foreign objects being mixed in	Update an existing standard	更换包装工作台面 Replace the packaging workbench surface		Heqin	Nov 2, 2025	Nov 2, 2025	N/A	N/A		N/A	N/A	N/A			
待焊接品摆放位置 Placement position of welded products	Update an existing standard	锡焊工作区域使用透明防护罩遮蔽，待焊产品放置在隔离罩外的待焊接区域 The soldering work area is covered with a transparent protective cover, and the products to be soldered are placed in the soldering area outside the isolation cover		Tim	Nov 6, 2025	Nov 6, 2025	YES	YES		N/A	N/A	N/A			
烙铁头清洁 Cleaning of soldering iron tip	Update an existing standard	更新手工锡焊作业指导书，烙铁头每焊接完一个产品需进行表面清洁 Update the manual soldering operation manual, and clean the surface of the soldering iron tip after soldering each product		Kerwin	Nov 3, 2025	Nov 3, 2025	YES	YES		N/A	N/A	N/A			
锡焊作业员手套清洁度 Cleanliness of soldering operator gloves	Update an existing standard	更新手工锡焊作业指导书，增加佩戴手套要求及手套更换频次 Update manual soldering operation manual, increase glove wearing requirements and glove replacement frequency		Kerwin	Nov 3, 2025	Nov 3, 2025	YES	YES		N/A	N/A	N/A			
锡焊操作步骤 Soldering operation steps	Update an existing standard	更新手工锡焊作业指导书，明确锡焊的具体操作步骤及注意事项 Update the manual soldering operation manual to clarify the specific steps and precautions for soldering		Kerwin	Nov 3, 2025	Nov 3, 2025	YES	YES		N/A	N/A	N/A			
锡焊手持电烙铁烙铁头位置 Position of soldering iron tip	Update an existing standard	更新手工锡焊作业指导书，增加对手持电烙铁的角度要求 Update the manual soldering operation manual to include angle requirements for handheld electrical wires		Kerwin	Nov 3, 2025	Nov 3, 2025	YES	YES		N/A	N/A	N/A			

ACTION PLAN FOR PERMANENT COUNTERMEASURES - NON DETECTION

ROOT-CAUSE [5] select the Root Cause among the list proposed	TYPE OF COUNTERMEASURE [6]	COUNTERMEASURE [7]	STANDARD NUMBER REVISION LEVEL [8]	WHO [9]	Due date [10]	Done date [11]	CHECK: Are ALL users including their Direct Manager trained to use this STANDARD? [12]	CHECK: Is there an ENFORCEMENT plan for the Direct manager to ensure the respect of this STANDARD? [13]	CHECK: Genba Check Done and OK (Date) [14]	ACTI: Document updated? Design FMEA [15]	ACTI: Document updated? Process FMEA [16]	ACTI: Document updated? Control Plan [17]	ACTI: Date when all relevant FMEAs and Control Plans were updated [18]	ACTI: Generalization to other lines and/or products within the site [19]	ACTI: LLC Done? YES /NO [20]
增加刷架表面、刷盒内腔洁净、通止规自检动作增加表面、刷盒内腔外观全检，并进行刷架、刷盒内腔外观全检。 Increase the surface of the brush holder, clean the inner cavity of the brush box, and perform self inspection of the on/off gauge	Update an existing standard	在装连接件工序中增加全检，作业员进行刷架板外观，刷盒内腔外观全检。 更新检验指导书，全检需使用两个止规，一体通规全检，止规止为OK Add a full inspection in the process of installing connectors, where the operator conducts a full inspection of the appearance of the brush grip plate and the interior of the brush box. Update the inspection manual to require the use of two stop gauges for full inspection. One stop gauge is used for full inspection, and the stop gauge is considered OK		Kerwin	Nov 7, 2025		YES	YES		N/A	N/A				
待检、GP12外观检验标准中无法识别卡顿失效 The appearance inspection standard cannot identify stuttering failure	Update an existing standard	待检、GP12外观检验标准中增加手动拨碳刷检测，为期3个月。 Add manual carbon brush inspection to the final inspection and GP12 appearance inspection standards for a period of 3 months.		Tim	Nov 11, 2025		YES	YES		N/A	N/A				
碳刷回弹性能测试采用导通方式检测，未有具体力值监控，若碳刷存在轻微卡滞，回弹力若存在误判风险 The test for the rebound performance of the carbon brush is conducted using the conduction method. There is no specific force value monitoring. If there is slight sticking of the carbon brush, there is a risk of incorrect judgment regarding the rebound force.	Update an existing standard	导通性能测试设备改善，调整PLC碳刷滞动延时时间，由0.08S调整为0.03S，可以提升碳刷回弹力小、卡顿异常的识别能力 Improvement of conductivity testing equipment, adjustment of PLC carbon brush sliding delay time from 0.08S to 0.03S, can enhance the recognition ability of carbon brush rebound force and abnormal jamming		Jim	Nov 10, 2025		YES	YES		N/A	N/A				